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RESEARCH ARTICLE

Graph-based feature learning methods for subject-dependent and subject-independent motor imagery EEG decoding

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Abstract

The significant intra-individual variability and inter-individual differences in scalp electroencephalogram (EEG) make it difficult to learn task-distinguishable features, posing a challenge for motor imagery brain-computer interfaces. Current feature learning methods often produce an incomplete feature space, struggling to accommodate these variations and differences. Additionally, the weak discriminative nature of this feature space results in diminished EEG classification performance. This paper introduces novel graph-based feature learning methods to improve motor imagery decoding performance in both subject-dependent and subject-independent contexts. Firstly, construct a complete time–frequency-spatial-graph (TFSG) feature space. The original EEG signals are segmented into multiple time–frequency units using filter banks and sliding time windows. Spatial and brain network-based graph features are then extracted from each time–frequency unit and fused to create the TFSG features. This fused feature space is larger and more inclusive, effectively accommodating both intra- and inter-individual EEG variations. Secondly, learn a discriminative TFSG feature space. Two advanced methods are proposed. The first method employs a nonconvex sparse optimization model with log function regularization, which reduces bias in model estimation, thereby enabling more accurate learning of EEG patterns. The second method incorporates Fisher's criterion regularization into a sparse optimization framework to improve feature separability. A unified algorithmic framework is developed to solve the two new models. Our methods are validated on two motor imagery EEG datasets, achieving the highest average classification accuracies of 82.93, 68.52, and 71.69% for subject-dependent, subject-independent, and subject-adaptive evaluation methods, respectively. Experimental results demonstrate that the developed TFSG features significantly enhance both subject-dependent and subject-independent decoding performance, while the proposed regularization models improve the discriminability of the feature space, leading to further advancements in motor imagery decoding performance.

Keywords Motor imagery · EEG decoding · Time–frequency-spatial-graph feature · Feature extraction · Feature selection

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Introduction

A Brain-Computer Interface (BCI) system establishes a direct communication channel between humans and computers by decoding brain signals and converting them into control commands for a computer or external device (Wu et al. 2024). Electroencephalography (EEG) is particularly advantageous for BCI systems due to its non-invasive nature and high temporal resolution, offering significant benefits in terms of physiological safety and real-time control (Meng et al. 2024). Motor imagery EEG is especially promising for motor control applications such as stroke rehabilitation and neuroprosthetic control. However, challenges such as low signal-to-noise ratio, high randomness in motor imagery EEG, and both intra-individual variability and inter-individual differences pose significant obstacles to effective motor imagery decoding (Altaheri et al. 2023). Improving the accuracy and stability of motor imagery EEG decoding remains a crucial challenge in advancing motor imagery BCI systems.

Effective feature learning methods, including feature extraction and selection, are crucial in EEG decoding, especially for motor imagery applications (Phadikar et al. 2023; Lu et al. 2024). Over the years, significant technical advancements have led to the development of numerous methods for feature extraction and selection in motor imagery decoding (Fu et al. 2023; Sharma et al. 2023). However, these methods still face certain limitations. This paper aims to address these challenges by presenting innovative solutions for feature extraction and selection in the motor imagery decoding field. The current research status and the issues with existing feature extraction and selection methods will first be reviewed. Subsequently, the proposed research ideas and solutions will be introduced.

In recent years, research on EEG time–frequency-spatial feature extraction has garnered increasing attention (Liu et al. 2022). These studies have shown that motor imagery is associated with specific frequency bands, time windows, and the spatial distribution of the cerebral motor cortex. Additionally, it is now understood that these aspects interact and influence each other, necessitating their simultaneous and joint optimization (Luo 2023). Consequently, extracting time–frequency-spatial features has become fundamental to the field of motor imagery decoding (Li et al. 2023).

However, existing time–frequency-spatial features still fall short in comprehensively characterizing EEG signal patterns. The human brain functions as a complex network, with multiple brain regions collaborating. During specific cognitive tasks, valuable information is intricately embedded within this complex brain network (Jin et al. 2021a). Graph features derived from brain networks have been applied in motor imagery EEG decoding (Ai et al. 2019; Shamsi et al. 2021), and research indicates that these features can enhance classification accuracy in motor imagery BCI contexts (Zhang et al. 2019). These graph features have also been used in biometric identification (Wang et al. 2020) and to interpret neural mechanisms (Leng et al. 2024), suggesting their potential to effectively characterize motor imagery EEG signal patterns.

To capture a more comprehensive set of EEG information, the simultaneous extraction of time–frequency-spatial-graph (TFSG) fused features is proposed. This approach allows for a more comprehensive characterization of EEG signal patterns, offering more effective feature information for subject-dependent decoding and creating a broader common feature space for subject-independent decoding. Consequently, the new method is better equipped to adapt to intra-individual variations and inter-individual differences in EEG signals. To the best of our knowledge, this particular approach has not been previously reported.

Selecting useful and discriminative feature information from the complete set of EEG features poses another critical challenge. In the realm of motor imagery EEG feature selection, the least absolute shrinkage and selection operator (LASSO) model is a commonly employed and effective technique (Miao et al. 2021; Zhang et al. 2022; Jin et al. 2021b; Meng et al. 2021; Sun et al. 2024). Compared to other filter and wrapper methods (Sadiq et al. 2021; Li et al. 2020a; Zheng et al. 2021), the LASSO method performs feature selection and classification simultaneously, simplifying data processing steps and offering higher decoding accuracy, thus providing distinct advantages. Additionally, the LASSO model can be effectively addressed by existing convex optimization techniques.

However, as technology evolves, the limitations of the LASSO model are becoming increasingly apparent. Firstly, the LASSO model is prone to biased estimation (Wen et al. 2018), lacks robust noise suppression capability (Wang et al. 2018), and has limitations in inducing sparsity (Zhang et al. 2020a). When applied to EEG decoding, these shortcomings can lead to imprecise feature selection and less robust feature classification. In contrast, nonconvex regularization techniques offer a significant advantage in addressing the biased estimation problem (Wen et al. 2018). Secondly, the LASSO model primarily focuses on selecting important features during feature selection without sufficiently considering feature separability. To address these issues, the LASSO model is enhanced by incorporating nonconvex and Fisher's criterion regularization. This enhancement will enable the learning of more accurate and separable EEG patterns, leading to a more discriminative TFSG feature space (Sun et al. 2023).

To address the aforementioned challenges, this paper presents graph-based feature learning methods for motor imagery EEG decoding. Firstly, construct a complete TFSG feature space. The original EEG signal is divided into 30 time–frequency units using filter banks and sliding time windows. Spatial features and brain network-based graph features are then extracted from each time–frequency unit. Finally, the spatial-graph features from the 30 time–frequency units are fused to generate the TFSG multi-domain features. Spatial features are extracted using the Common Spatial Pattern (CSP) method (Gaur et al. 2021). For rich graph information, three brain network construction methods are combined with eight types of graph features. Secondly, learn a discriminative TFSG feature space. Two discriminative feature selection models incorporating nonconvex and Fisher's criterion regularization are proposed. One model addresses the biased estimation problem of LASSO, while the other improves the feature separability of LASSO. A unified algorithmic framework is designed to solve these models. The effectiveness of the proposed method is validated on two publicly available motor imagery EEG datasets using three evaluation methods: subject-dependent, subject-independent, and subject-adaptive decoding.

The main contributions and innovations of this paper are summarized as follows:

(1) Comprehensive Feature Fusion Method: A method for time–frequency-spatial-graph multi-domain feature fusion is proposed, which aims to extract more comprehensive EEG features and provide a thorough characterization of EEG signal patterns. This method is better suited to adapt to intra-individual variations and inter-individual differences in EEG signals, leading to superior classification results in both subject-dependent and subject-independent decoding.

(2) Discriminative Feature Selection Methods: Two discriminative feature selection methods, leveraging nonconvex and Fisher's criterion regularization, are introduced to acquire more precise and separable EEG information. These methods are designed to simultaneously conduct feature selection and classification, eliminating the need for additional classifiers. The proposed methods significantly improve the discriminability of the feature space, leading to substantial enhancements in classification performance and a reduction in model training time.

(3) Unified Algorithmic Framework: The LASSO model and the two newly proposed models can be expressed in a unified form. A unified algorithmic solution framework for these models is designed, enabling the uniform solution of sparse selection models with different regularization terms such as convex optimization, nonconvex optimization, and Fisher's criterion.

(4) Extensive Experimental Validation: To evaluate the effectiveness of the proposed method, a comprehensive set of experiments was carried out, employing three evaluation approaches: subject—dependent, subject—independent, and subject—adaptive. Furthermore, a detailed, systematic analysis of the proposed feature extraction and selection methods was provided, presented, and discussed in detail.

The structure of this paper is as follows: Section II provides an overview of the experimental data; Section III introduces the TFSG feature extraction method, as well as the feature selection and classification methods. Section IV presents and analyzes the experimental results. Discussions are outlined in Section V, and conclusions are provided in Section VI.

Data description

The proposed methods are validated using two publicly available motor imagery EEG datasets: Dataset 1 sourced from the international BCI competition and Dataset 2 obtained from the BNCI Horizon 2020 project.

Dataset 1: Dataset IIa from BCI Competition IV (Zhang et al. 2020b). This dataset comprises nine healthy subjects, denoted as A01, A02, …, A09. Each subject performs four types of motor imagery tasks: left-hand, right-hand, foot, and tongue. Since we focus on binary classification tasks, these four types of tasks are recombined to form six sets of binary classification tasks. The dataset utilizes 22 electrode channels with a sampling rate of 250 Hz. The division of the training and test sets for each subject adheres to the BCI competition criteria, with 72 samples for each type of task in both sets. For further details, please refer to the official website: https://www.bbci.de/competition/iv/.

Dataset 2: Database of the BNCI Horizon 2020 project, no. 002–2014 (Steyrl et al. 2016). This dataset includes 14 healthy subjects, identified as S01, S02, …, S14. Each subject performs two types of motor imagery tasks: right-hand and foot. The dataset employs 15 electrode channels with a sampling rate of 512 Hz. To reduce the amount of data and calculations, the data were downsampled to 256 Hz in this paper. Each subject participated in 8 rounds of the experiment, consisting of 20 trials per round (10 trials per task). The data from the first five rounds are used as the training set, while the data from the last three rounds are used as the test set. Consequently, the training and test sets for each subject amount to 100 and 60, respectively. For additional information, please refer to the official website: http://bnci-horizon-2020.eu/database/data-sets.

The proposed methods

The algorithm framework and data processing flow of the proposed methods are illustrated in Fig. 1, comprising three main steps.

First, Division of Time–Frequency Units (Liu et al. 2022; Luo 2023): The original EEG signal undergoes bandpass filtering using a filter bank with bands ranging from 8–12 to 26–30 Hz. Each subband signal is segmented using sliding time windows of 0.5–2.5 s, 1–3 s, and 1.5–3.5 s to extract single-trial data. This results in a total of 10 frequency bands and 3 time windows, creating 30 time–frequency units. Each time–frequency unit represents single-trial data, aligned with the labels of the original EEG signal.

Second, Feature Extraction: Fig. 1 depicts two feature extraction methods: traditional time–frequency-spatial features and the newly proposed time–frequency-spatial-graph features. The first method involves extracting spatial (S) features from multiple time–frequency units (MTF), referred to as the MTF-S method. The second method extracts spatial-graph (SG) fusion features from multiple time–frequency units (MTF), known as the MTF-SG method. In the MTF-SG method, graph features are derived from the brain network-based weighted adjacency matrix. Detailed descriptions of these two feature extraction methods will be provided in subsequent sections.

Third, Feature Selection and Classification: Alongside the commonly used LASSO method, two new methods are proposed for simultaneous feature selection and classification of the extracted features.

This structured approach aims to enhance the characterization of EEG signal patterns and improve classification accuracy in motor imagery decoding tasks. In Fig. 1, brain network construction methods such as PLV, PLI, and WPLI are represented. These methods contribute to the formation of brain network-based weighted adjacency matrices. Graph feature calculation methods including DE, BC, and TR are utilized to derive features from these networks. The feature selection and classification methods are denoted by LASSO, LOG, and LASSO-F. Detailed introductions to these methods will be provided in the subsequent sections.

Feature extraction methods

This section begins with the presentation of the conventional time–frequency-spatial feature extraction method (MTF-S), followed by an introduction to the proposed time–frequency-spatial-graph feature extraction method (MTF-SG).

The MTF-S feature extraction method

The MTF-S method initially decomposes the original EEG data into multiple time–frequency units. Subsequently, CSP features are extracted from each time–frequency unit, and the resulting CSP features from multiple time–frequency units are concatenated into a feature vector.

(1) CSP feature extraction.

The objective function of the CSP method is formulated as follows (Blankertz et al. 2008):

$$J(\mathbf{w}_S) = \frac{\mathbf{w}_S^T \overline{\mathbf{C}_1} \mathbf{w}_S}{\mathbf{w}_S^T \overline{\mathbf{C}_2} \mathbf{w}_S} \tag{1}$$

where $\mathbf{w}_S \in R^C$ denotes the spatial filter, C denotes the number of EEG electrode channels, and $\overline{\mathbf{C}_i}$ denotes the average covariance matrix of the task i.

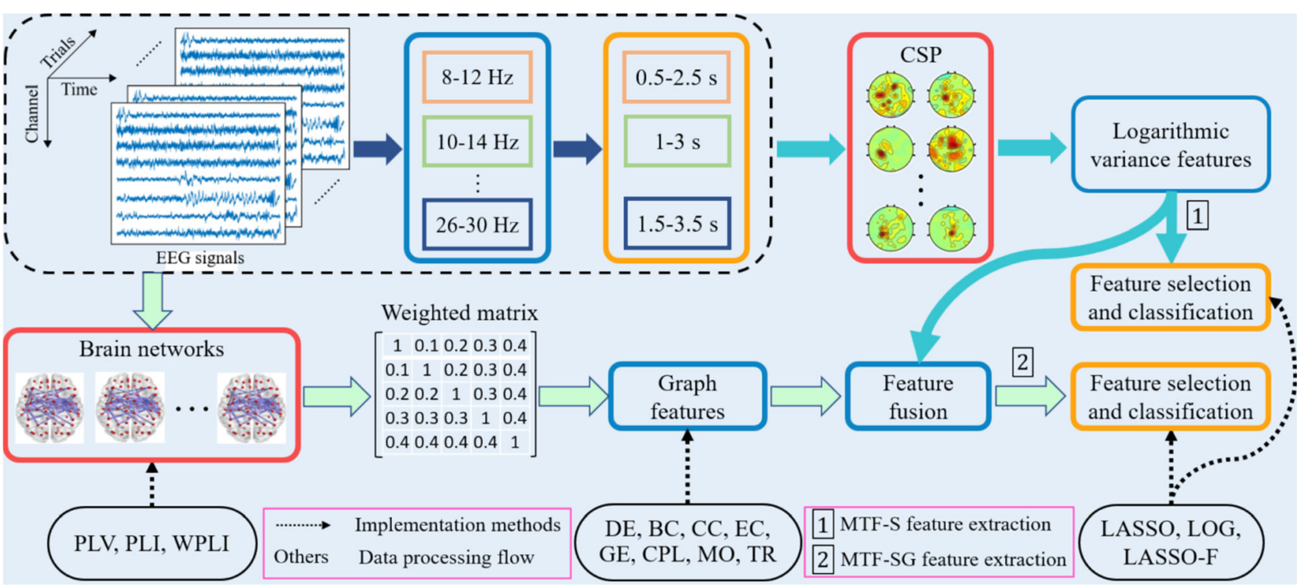

Fig. 1 The algorithm framework and data processing flow of the proposed methods. The dashed box outlines the division method for multiple time–frequency units. Solid arrows indicate the sequential data processing steps, while dashed arrows denote the implementation method for each processing step. Specifically, blue-green solid arrows represent the data processing steps for the MTF-S feature extraction method, while light-green solid arrows signify the data processing steps for the MTF-SG feature extraction method

Equation (3–1) can be reformulated into a generalized eigenvalue problem for solving (Lotte and Guan 2011), and the resulting eigenvectors serve as the sought-after spatial filters. Subsequently, the eigenvectors corresponding to the first m maximum eigenvalues and the last m minimum eigenvalues are selected to construct the final spatial filters $\mathbf{W}$, $\mathbf{W} \in R^{C\times 2\mathrm{m}}$; m represents the pair number of spatial filters, which is set to 1 in this paper.

For single-trial data $\mathbf{D}$, the spatially filtered signals are as follows:

$$\mathbf{Z} = \mathbf{W}^T\mathbf{D} \tag{2}$$

where ^{T} indicates the transpose operation. $\mathbf{Z} \in R^{2m\times T}$, $\mathbf{D} \in R^{C\times T}$, C denotes the number of electrode channels, and T denotes the number of samples per electrode channel. The features of the single-trial data $\mathbf{D}$ can be further derived as follows:

$$s_p = \log\left(\frac{\mathrm{var}(\mathbf{Z}_p)}{\sum_{i=1}^{2m}\mathrm{var}(\mathbf{Z}_i)}\right), p = 1, 2, \cdots, 2m \tag{3}$$

where $\log(\cdot)$ denotes logarithmic operation and $\mathrm{var}(\cdot)$ denotes variance operation, $\mathbf{Z}_p$ denotes the p-th channel signal of $\mathbf{Z}$, $\mathbf{Z}_i$ is similar.

(2) Spatial features of multiple time–frequency units.

Applying the computational rules outlined above, a set of CSP log-variance features is extracted for each time–frequency unit. The spatial features of the 30 time–frequency units are concatenated as follows:

$$\mathbf{s} = [s_1^1, s_2^1, \cdots, s_{2m}^1; s_1^2, s_2^2, \cdots, s_{2m}^2; \cdots\cdots; s_1^{30}, s_2^{30}, \cdots, s_{2m}^{30}] \tag{4}$$

where $\mathbf{s}$ represents the traditional time–frequency-spatial feature vector, $s_1^1, s_2^1, \cdots, s_{2m}^1$ denotes the spatial features of the first time–frequency unit, and the others are similar.

The proposed MTF-SG feature extraction method

The MTF-SG method initially extracts the spatial and grap features from each time–frequency unit, respectively. Subsequently, spatial-graph features are fused for each time–frequency unit, and the resulting fused spatial-graph features from multiple time–frequency units are concatenated into a feature vector.

(1) Graph features of multiple time–frequency units.

The method for extracting graph features based on brain networks primarily consists of two sequential steps: initial establishment of the brain network and subsequent computation of graph features. Let's delve into each step individually.

(a) Brain network construction.

As depicted in Fig. 2, the construction of the brain network occurs at each time–frequency unit using single-trial data. In this process, each electrode channel from the EEG signal acts as a node in the graph. The next step involves selecting appropriate metrics to evaluate the relationships between pairs of electrode channels, which then form the edges of the graph. Subsequently, nodes and edges are organized into a weighted adjacency matrix. It's important to note that the brain network is established based on functional connectivity, without considering the direction of information flow between electrodes. Therefore, the resulting weighted adjacency matrices are undirected and symmetric.

During the brain network construction phase, a crucial aspect is selecting metrics that effectively capture the interactions between electrode signals. For establishing brain networks, three commonly used metrics based on phase synchronization were chosen: phase locking value (PLV), phase lag index (PLI), and weighted phase lag index (WPLI). The formulas for calculating PLV, PLI, and WPLI can be found in the literature (EEG 2019).

For EEG signals with C electrode channels, a weighted adjacency matrix of size $C \times C$ is derived. Utilizing this weighted matrix, graph features for each time–frequency unit are then computed.

(b) Graph feature calculation.

Utilizing the weighted adjacency matrix, graph theory is applied for the analysis and computation of graph features in the brain network. Specifically, four global features and four local features are incorporated. For global features, a single feature is extracted per time–frequency unit. In contrast, local features align with the number of electrode channels, where each feature corresponds to an individual electrode channel (graph node). Global features include characteristic path length (CPL), global efficiency (GE), modularity (MO), and transferability (TR). The local features encompass degree (DE), betweenness centrality (BC), clustering coefficients (CC), and eigenvector centrality (EC). The calculation formulas for these eight types of graph features are elucidated in the literature [15 and 35]. All computations of graph features are performed using the brain connectivity toolbox (Rubinov and Sporns 2010) (http://www.brainconnectivitytoolbox.net/).

In the case of local graph features, the final feature is derived by calculating the average of all node features. Consequently, for both local and global features, only one feature is extracted per time–frequency unit. The graph features obtained from 30 time–frequency units are concatenated in the following manner.

$$\mathbf{g} = [g^1, g^2, \cdots, g^{30}] \tag{5}$$

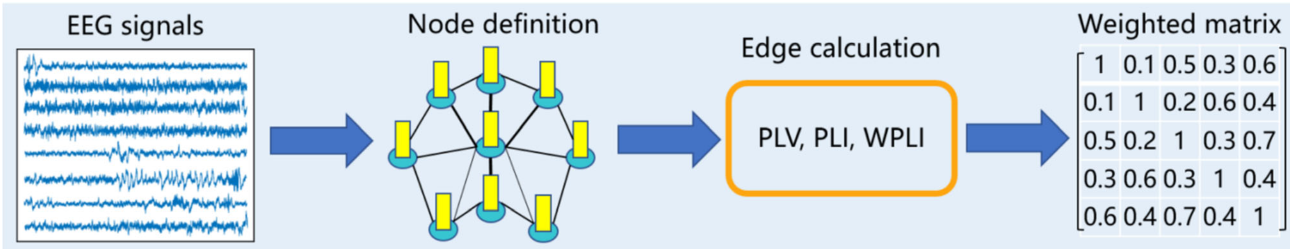

Fig. 2 Brain network construction

where g denotes one type of local or global graph feature. $\mathbf{g}$ denotes the time–frequency-graph feature vector, which will be integrated with the previously mentioned time–frequency-spatial feature vector.

(2) Spatial-graph feature fusion of multiple time–frequency units.

The fusion of the extracted time–frequency-spatial and time–frequency-graph features is illustrated in Fig. 3. It is important to highlight that there are eight types of graph features, and the experimental workload becomes substantial when combining various types. Additionally, through experimentation, we have observed that the classification performance does not exhibit significant improvement when combining more than three types of graph features. Consequently, we limit the combination to a maximum of two types of graph features, ensuring that one type comprises global graph features while the other encompasses local graph features.

In the fusion of spatial-graph features with a single type of graph feature, one of the phase synchronization metrics is selected to construct the adjacency matrix, resulting in the computation of only one type of graph feature. For the fusion of spatial-graph features with two types of graph features, the process is similar to that of a single type of graph feature. However, in this case, both a local feature and a global feature are computed and combined during the extraction of graph features.

Feature selection and classification methods

The extracted multi-domain features, which include time, frequency, spatial, and graph information, provide a comprehensive representation of EEG patterns, integrating rich EEG information. However, within this extensive feature set, redundancy may exist. In subject-dependent decoding, it is essential to pinpoint and select EEG patterns specific to each subject from this comprehensive set of EEG features. Conversely, for subject-independent decoding, the goal is to establish a larger common feature space. Therefore, feature selection plays a crucial role in optimizing the relevance and efficiency of features utilized in decoding processes.

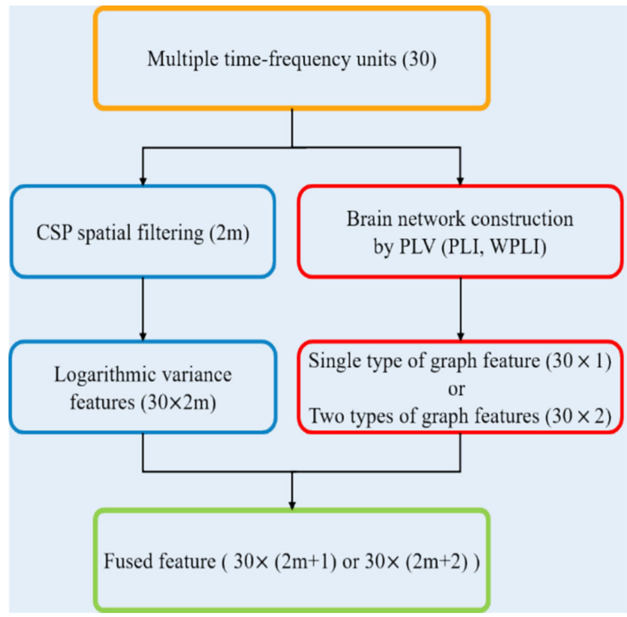

Fig. 3 Spatial-graph feature fusion of multiple time–frequency units

This paper introduces embedded feature selection methods based on sparse optimization models to streamline data processing steps in EEG decoding. Initially, the well-established LASSO model, known for its convex sparse optimization properties, is outlined. Following this, two novel sparse optimization models are introduced: LOG (model weights regularized using the log function) and LASSO-F (LASSO model enhanced with Fisher's criterion regularization). To consolidate these methods, a unified algorithmic framework is designed for solving these models.

The LASSO model

The formulation of the LASSO model is provided as follows (Zhang et al. 2021):

$$\mathbf{w} = \arg\min_{\mathbf{w}} \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_1 \tag{6}$$

where $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N)^T \in R^{N\times P}$ is the sample matrix of fused features, N is the number of samples, and P is the feature dimension. $\mathbf{w} \in R^P$ is the model coefficient, which also corresponds to the weights of the features.$\mathbf{y} = [y_1, y_2, \cdots, y_N]^{\mathrm{T}}$ is the sample label, $y_i \in \{-1, 1\}$. $\|\cdot\|_2^2$ denotes the square of the L_2 norm, $\|\cdot\|_1$ denotes the L_1 norm, and $\lambda > 0$ is the regularization parameter.

The core principle of the feature selection model based on sparse optimization is to induce as many variables in the weight coefficients, represented by $\mathbf{w}$, to be zero through regularization. Theoretically, the L_0 norm is the ideal choice (Wen et al. 2018). However, solving the regularization based on the L_0 norm is exceptionally challenging. LASSO employs the L_1 norm as an approximation to the L_0 norm, making it a biased estimation model (Wen et al. 2018). Additionally, the features selected with LASSO tend to be excessively sparse and scattered. In scenarios where the number of feature dimensions significantly exceeds the number of samples (a common occurrence in EEG decoding), the resulting features of LASSO may become unstable. Therefore, there is a need to enhance the robustness of the LASSO model.

The proposed LOG model

The objective function of the LOG model is expressed as follows:

$$\mathbf{w} = \underset{\mathbf{w}}{\arg\min} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \log\left(1 + \frac{\|\mathbf{w}\|_1}{a}\right) \tag{7}$$

where $a > 0$ represents the scaling parameter and the remaining terms are consistent with the LASSO model. In this paper, a is set to 0.02. In comparison to the LASSO model (with L_1 norm regularization), the nonconvex LOG model, utilizing a log function as the penalty term, exhibits superior capability in inducing sparsity compared to the L_1 norm. Moreover, it unevenly penalizes all elements, a characteristic that makes the log function closer to the L_0 norm than the L_1 norm, resulting in an approximately unbiased estimation (Zhang et al. 2020a). In contrast, the LASSO model penalizes all components of the weight coefficient vector equally. While it compresses the weight coefficients associated with noise features to zero, it also compresses the weight coefficients related to target features to some extent, leading to a biased estimation of the weight coefficients for the target features.

The proposed LASSO-F model

Fisher's criterion identifies a projection vector $\mathbf{w}_F$ such that, upon projecting the two types of samples using $\mathbf{w}_F$, the distance between samples of the same class is minimized, while the distance between samples of different classes is maximized. This objective is achieved by maximizing the following criterion function.

$$J(\mathbf{w}_F) = \frac{\mathbf{w}_F^T \mathbf{S}_B \mathbf{w}_F}{\mathbf{w}_F^T \mathbf{S}_W \mathbf{w}_F} \tag{8}$$

where $\mathbf{S}_B$ and $\mathbf{S}_W$ denote the between-class scatter matrix and within-class scatter matrix of the original sample matrix $\mathbf{X}$, respectively (Hoffmann et al. 2008).

Maximizing $J(\mathbf{w}_F)$ is equivalent to maximizing $\mathbf{w}_F^T \mathbf{S}_B \mathbf{w}_F$ while minimizing $\mathbf{w}_F^T \mathbf{S}_W \mathbf{w}_F$, and maximizing $\mathbf{w}_F^T \mathbf{S}_B \mathbf{w}_F$ is equivalent to minimizing $-\mathbf{w}_F^T \mathbf{S}_B \mathbf{w}_F$. Therefore, maximizing $J(\mathbf{w}_F)$ can be reformulated as minimizing the following form.

$$R(\mathbf{w}_F) = (\mathbf{w}_F^T \mathbf{S}_W \mathbf{w}_F - \mathbf{w}_F^T \mathbf{S}_B \mathbf{w}_F) \tag{9}$$

After projecting the feature samples with $\mathbf{w}_F$, their feature separability is enhanced. Building upon this concept, we incorporate $R(\mathbf{w}_F)$ as a new regular term for the LASSO model, taking into account feature separability during the selection of significant features. Denoting $\mathbf{w}_F$ uniformly as $\mathbf{w}$, the objective function of the LASSO-F model is expressed as follows:

$$\mathbf{w} = \underset{\mathbf{w}}{\arg\min} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_1 + \rho R(\mathbf{w}) \tag{10}$$

We assign equal importance to feature separability and label loss, setting ρ to 1 in this paper. It is noteworthy that models incorporating Fisher's criterion regularization do not introduce an increase in the number of model parameters.

For intuitive comparison, the parameters of the above three models are shown in Table 1.

The proposed unified model-solving framework

Models (3–6)–(3–7), and (3–10) can be collectively expressed as a composite optimization problem with the following form.

$$\min_{\mathbf{w}} \; g(\mathbf{w}) + h(\mathbf{w}) \tag{11}$$

where $g(\mathbf{w})$ represents the smooth part of the function, such as $g(\mathbf{w}) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$ or $g(\mathbf{w}) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + R(\mathbf{w})$. $h(\mathbf{w})$ is the regular term, a non-smooth function employed to ensure specific structural information during problem-solving, such as L_1 norm and LOG penalties.

The proximal gradient method will be employed to solve this type of problem. The proximal operator serves as a highly effective tool for addressing non-smooth problems and is closely linked to the design of various algorithms, including the proximal gradient method that will be introduced shortly.

The proximal operator is defined as:

$$\mathrm{prox}_h(\mathbf{u}) = \arg\min_{\mathbf{w}} \left\{ h(\mathbf{w}) + \frac{1}{2}\|\mathbf{w}-\mathbf{u}\|_2^2 \right\} \tag{12}$$

It can be seen that the objective of the proximal operator is to solve for a point that is not too far from the point $\mathbf{u}$ and to make the function value $h(\mathbf{w})$ also relatively small. Several common examples are provided below, where the computation of the proximity operator essentially involves solving an optimization problem.

When $h(\mathbf{w}) = \lambda\|\mathbf{w}\|_1$,

$$\mathrm{prox}_h(\mathbf{u}) = \begin{cases} \mathbf{u}-\lambda t, & \mathbf{u} > \lambda t, \\ 0, & |\mathbf{u}| \le \lambda t, \\ \mathbf{u}+\lambda t & \mathbf{u} < -\lambda t. \end{cases} \tag{13}$$

where $t > 0$ is the step size.

When $h(\mathbf{w}) = \lambda \log(1 + \frac{\|\mathbf{w}\|_1}{a})$,

$$prox_h(\mathbf{u}) = \frac{sign(\mathbf{u})}{2}\left(|\mathbf{u}| - a + \sqrt{(a-|\mathbf{u}|)^2 + 4\max(a|\mathbf{u}| - \lambda t, 0)}\right) \tag{14}$$

Built upon the proximal operator, the proximal gradient method consists of two fundamental steps.

(1) Gradient descent step.

Given an iteration point $\mathbf{w}^k$, define an intermediate point $\mathbf{v}^k$, such that:

$$\mathbf{v}^k = \mathbf{w}^k - t\nabla g(\mathbf{w}^k) \tag{15}$$

where $\nabla g(\mathbf{w}^k)$ is the gradient of $g(\mathbf{w})$ at the point $\mathbf{w}^k$.

$$\begin{gathered}\nabla g(\mathbf{w}^k) = \mathbf{X}^T(\mathbf{X}\mathbf{w}^k - \mathbf{y}) \;\; or \\ \nabla g(\mathbf{w}^k) = \mathbf{X}^T(\mathbf{X}\mathbf{w}^k - \mathbf{y}) + 2(S_W\mathbf{w}^k - S_B\mathbf{w}^k)\end{gathered} \tag{16}$$

That is, the initial step involves performing a gradient descent.

(2) Proximal operator step.

Calculate the proximal operator of the non-differentiable function $h(\mathbf{w})$, i.e.

$$\mathbf{w}^{k+1} = prox_h(\mathbf{v}^k) = prox_h(\mathbf{w}^k - t\nabla g(\mathbf{w}^k)) \tag{17}$$

In summary, the proximal gradient method essentially involves linearly expanding the smooth part of the problem, introducing a quadratic term, and retaining the non-smooth part. It then seeks the minimum as the estimate at each step. The first step in the solution corresponds to gradient descent, while the second step involves contraction. The operator in the second step, often referred to as the shrinkage operator, ensures the sparse structure of the solution throughout the iterative process.

Experiments

Evaluation methods and parameter setting

In this paper, the evaluation criterion for assessing performance is the classification accuracy on the test set. The proposed methods are evaluated using three approaches: subject-dependent decoding, subject-independent decoding, and subject-adaptive decoding.

In subject-dependent decoding, a separate model is trained for each subject. The division of the training and test sets for each model aligns with the specifications outlined in the Data Description section.

In subject-independent decoding, the model is trained using data from all subjects except the target subject. The test set of the model comprises the test set of the target subject, while the training set of the model incorporates all training and test sets from the other subjects. To illustrate,

Table 1 A summary of parameters for different models

Models	Parameters
LASSO	$\mathbf{w}$,λ
LOG	$\mathbf{w}$,λ,a
LASSO-F	$\mathbf{w}$,λ,ρ

Table 2 Average classification accuracy of subject-dependent decoding (unit: %)

Datasets	MTF-S (Baseline)			MTF-SG (PLV + DE)		
	LASSO (Baseline)	LOG	LASSO-F	LASSO	LOG	LASSO-F
Dataset 1(LvsR)	76.54	**79.55**	79.01	77.78	**79.55**	**79.55**
Dataset 1(LvsF)	86.57	88.27	87.81	87.11	**89.58**	88.89
Dataset 1(LvsT)	84.18	86.81	86.88	85.57	**88.66**	88.43
Dataset 1(RvsF)	82.49	83.64	**84.10**	82.25	83.03	82.02
Dataset 1(RvsT)	80.25	82.56	81.94	83.18	**85.18**	84.49
Dataset 1(FvsT)	76.70	78.32	77.47	76.00	**78.55**	78.39
Dataset 2	73.69	77.38	76.55	74.17	**78.45**	78.33
Mean	79.59	81.99	81.57	80.37	**82.93**	82.54

Bold text indicates the method with the highest classification accuracy in the target dataset if subject 1 in dataset 1 is designated as the target subject, the training set for the model consists of all training and test sets from the other 8 subjects, and the test set includes the data from subject 1.

In subject-adaptive decoding, a pre-trained subject-independent model is utilized, keeping the regularization parameters λ of the model unchanged. The model is then retrained using the training set of the target subject, updating the weight coefficients $\mathbf{w}$ in the model. For instance, if subject 1 in dataset 1 is chosen as the target subject, a pre-trained model is trained using all the training and test sets of the other 8 subjects. The test set of subject 1 forms the test set for the model, and the training set of subject 1 is employed to fine-tune the pre-trained model by updating the weight coefficients $\mathbf{w}$.

To summarize, the test sets for all methods are consistent, with variations occurring only in the training process and data for model training. It's essential to note that none of the test sets are utilized in the training of the models in this paper.

The optimal regularization parameters of the three feature selection and classification models, i.e., the optimal values λ^*, are determined through tenfold cross-validation and grid search methods. The alternative set of regularization parameters is $\{2^{-5}, 2^{-4.8}, \cdots, 2^{4.8}, 2^5\}$.

Experimental results

Compared with the baseline method: MTF-S + LASSO

The average classification accuracy for subject-dependent, subject-independent, and subject-adaptive decoding is presented in Tables 2, 3 and 4. The highest classification accuracy is highlighted in bold. It's noteworthy that the MTF-SG method encompasses various approaches obtained by combining different brain network construction methods and various types of graph features. These details will be further discussed in the following sections. For comparison purposes, we select the MTF-SG method with the best classification results. In Tables 2, 3 and 4, MTF-SG(PLV + DE) refers to a time–frequency-spatial-graph feature extraction method based on PLV-based brain networks and node-degree graph features. The highest average classification accuracy achieved is 82.93%, 68.52%, and 71.69% for subject-dependent, subject-independent, and subject-adaptive decoding, respectively. Compared to the baseline method (MTF-S + LASSO), the

Table 3 Average classification accuracy of subject-independent decoding (unit: %)

Datasets	MTF-S (Baseline)			MTF-SG (PLV + DE)		
	LASSO (Baseline)	LOG	LASSO-F	LASSO	LOG	LASSO-F
Dataset 1(LvsR)	68.13	69.60	69.60	71.37	70.76	**71.45**
Dataset 1(LvsF)	63.27	67.67	66.67	69.37	**71.45**	70.45
Dataset 1(LvsT)	63.20	62.65	63.42	68.75	**71.99**	**71.99**
Dataset 1(RvsF)	65.35	64.20	64.27	**67.75**	67.28	67.21
Dataset 1(RvsT)	63.97	64.20	64.12	68.21	**69.98**	69.83
Dataset 1(FvsT)	56.33	58.18	58.42	58.56	59.57	**59.72**
Dataset 2	65.48	63.93	64.05	**69.76**	68.57	68.33
Mean	63.81	64.32	64.34	67.83	**68.52**	68.42

Bold text indicates the method with the highest classification accuracy in the target dataset

Table 4 Average classification accuracy of subject-adaptive decoding (unit: %)

Datasets	MTF-S (Baseline)			MTF-SG (PLV + DE)		
	LASSO (Baseline)	LOG	LASSO-F	LASSO	LOG	LASSO-F
Dataset 1(LvsR)	72.68	71.68	71.84	73.84	74.30	**74.31**
Dataset 1(LvsF)	69.06	70.91	70.37	71.92	72.45	**73.61**
Dataset 1(LvsT)	70.99	67.59	69.13	72.99	74.54	**75.08**
Dataset 1(RvsF)	70.76	68.36	68.05	72.30	**72.38**	72.30
Dataset 1(RvsT)	68.44	68.90	69.21	71.30	**72.84**	72.61
Dataset 1(FvsT)	61.27	60.19	60.18	62.96	63.50	**63.58**
Dataset 2	69.52	66.55	66.55	**72.14**	71.43	70.83
Mean	69.00	67.65	67.81	71.14	71.62	**71.69**

Bold text indicates the method with the highest classification accuracy in the target dataset

Table 5 Comparison of the classification accuracy with existing feature extraction methods (unit: %)

Datasets	CSP + FLDA	TRCSP + FLDA	FBCSP + LOG	FBCSP + LASSO-F	STFSCSP + LOG	STFSCSP + LASSO-F	MTF-SG(PLV + DE) + LOG	MTF-SG(PLV + DE) + LASSO-F
Dataset 1(LvsR)	78.16	78.94	78.78	79.86	83.33	**81.79**	79.55	79.55
Dataset 1(LvsF)	81.71	83.87	82.25	82.33	84.49	86.27	**89.58**	88.89
Dataset 1(LvsT)	83.41	85.19	81.79	80.40	86.65	86.50	**88.66**	88.43
Dataset 1(RvsF)	78.55	81.48	82.87	82.48	**88.73**	87.89	83.03	82.02
Dataset 1(RvsT)	79.09	83.33	81.79	81.71	**86.81**	86.42	85.18	84.49
Dataset 1(FvsT)	72.37	70.76	72.92	71.76	**80.63**	79.24	78.55	78.39
Dataset 2	75.83	75.12	72.02	71.19	76.07	74.05	**78.45**	78.33
Mean	78.26	79.47	78.41	77.99	**83.25**	82.49	82.93	82.54

Bold text indicates the method with the highest classification accuracy in the target dataset

average classification accuracy demonstrates improvements of 3.34%, 4.71%, and 2.69%, respectively.

Based on the classification results in Tables 2, 3 and 4, several conclusions can be drawn. Firstly, the MTF-SG method consistently outperforms MTF-S, regardless of the chosen feature selection and classification method. Secondly, across different feature extraction methods, the two proposed methods consistently outperform the LASSO method. Thirdly, when comparing feature selection and classification methods, the LOG method shows superior classification results in subject-dependent and subject-independent decoding. Conversely, the LASSO-F method demonstrates better classification results in subject-adaptive decoding.

Tables 2, 3 and 4 provide the average classification accuracy at the dataset level. For detailed classification accuracy for each subject, please refer to Tables S1 to S6 in the Supplementary file. Additionally, to facilitate visual comparison, the distribution of classification accuracy for the three assessment methods (subject-dependent, subject-independent, and subject-adaptive decoding) is presented in Figs. S1 to S3 of the Supplementary file.

Compared with other existing methods

To comprehensively evaluate the proposed methods, comparisons with existing feature extraction and selection methods were conducted using subject-dependent decoding evaluation.

Table 5 presents the comparison results of feature extraction methods, where four existing methods are evaluated. The CSP method (Lotte and Guan 2011) involves band-pass filtering raw EEG data from 8–30 Hz, extracting features using CSP within a 0.5–2.5 s time window, and employing FLDA (Hoffmann et al. 2008) for classification. The TRCSP method (Lotte and Guan 2011) uses the same preprocessing and classification methods as CSP but applies TRCSP for feature extraction. The FBCSP method (Ang et al. 2008) utilizes LOG-F and LASSO-F for feature selection and classification, with a time window of 0.5–2.5 s. The STFSCSP method (Miao et al. 2017) excludes channel selection, employing LOG-F and LASSO-F for feature selection and classification, aligned with the original literature. From Table 5, it is evident that MTF-SG(PLV + DE) outperforms all methods except for STFSCSP with LOG feature selection.

Table 6 presents the comparison results of feature selection methods, evaluating a total of seven methods, all using the MTF-SG(PLV + DE) feature extraction method. The Fisher's score (F-score) (Tang et al. 2022), neighbor component analysis (NCA) (Yang et al. 2012), slime mould algorithm (SMA) (Li et al. 2020b), and whale optimization algorithm (WOA) (Mirjalili and Lewis 2016) employ

Table 6 Comparison of the classification accuracy with existing feature selection methods (unit: %)

Datasets	F-score	NCA	SMA	WOA	LASSO	LOG	LASSO-F
Dataset 1(LvsR)	75.00	76.77	74.07	74.07	77.78	**79.55**	**79.55**
Dataset 1(LvsF)	86.27	86.81	80.40	78.78	87.11	**89.58**	88.89
Dataset 1(LvsT)	84.65	84.41	79.09	79.86	85.57	**88.66**	88.43
Dataset 1(RvsF)	81.02	82.56	80.56	80.25	82.25	**83.03**	82.02
Dataset 1(RvsT)	80.48	81.33	78.16	80.79	83.18	**85.18**	84.49
Dataset 1(FvsT)	**78.78**	77.47	69.06	70.68	76.00	78.55	78.39
Dataset 2	70.95	73.33	73.45	75.12	74.17	**78.45**	78.33
Mean	78.96	79.86	76.18	76.93	80.37	**82.93**	82.54

Bold text indicates the method with the highest classification accuracy in the target dataset

FLDA as the classification function. For NCA, the maximum iteration number was set to 1000, with the iteration step size determined using Wolf line search, which has a maximum iteration number set to 20. For SMA, the maximum iteration number was set to 100, with the ratio of randomly distributed mucilage individuals to the total value set to 0.03. For WOA, the maximum iteration number was set to 100, and the constant of the logarithmic helix shape was set to 1. From Table 6, it is evident that the classification results of LOG and LASSO-F are significantly superior to those of the existing feature selection methods.

For subject-dependent and subject-independent decoding evaluations, the proposed methods yielding the best classification results (as detailed in Tables 2 and 3) are compared with several prominent deep learning methods: Deep ConvNet (Schirrmeister et al. 2017), EEGNet-8,2 (Lawhern et al. 2018), Spectral-Spatial CNN (Kwon et al. 2020), and MIN2NET (Autthasan et al. 2021). The summarized results are presented in Table 7. Due to the substantial workload involved, we did not reproduce these deep learning methods and instead directly cited the results provided in the MIN2NET literature (Autthasan et al. 2021). Additionally, since these methods only report classification results for the left-hand vs right-hand (L vs R) binary classification task on dataset 1, we present our method's corresponding classification results for a direct comparison. As shown in Table 7, our method achieves superior classification performance.

Discussions

Initially, the comprehensive experimental results are delved into to elucidate key findings and offer insights for subsequent researchers. Subsequently, further experimental investigations and discussions concerning feature extraction methods, as well as feature selection and classification methods, are conducted, aiming to enhance the understanding of the proposed techniques. Finally, new avenues for future research are explored.

Overall classification results

Analyzing the classification results presented in Tables 2, 3 and 4 yields several distinct conclusions. Firstly, the classification performance of TFSG features surpasses that of time–frequency-spatial features, particularly evident in the subject-independent and subject-adaptive assessment methods. Secondly, the two proposed feature selection and classification methods exhibit significant improvements over LASSO, underscoring the effectiveness of non-convex and Fisher's criterion regularization in enhancing LASSO.

Analyzing the classification results from Tables 5 and 6 reveals that the proposed methods outshine existing feature extraction and feature selection approaches. Our proposed sparse selection model is capable of concurrently performing feature selection and classification, leading to enhanced classification performance and reduced data processing steps compared to filtered and wrapped feature selection methods. In Table 7, the classification results indicate that our proposed method significantly outperforms several existing deep learning methods in both subject-dependent and subject-independent assessments. Beyond the improvement in classification performance, our method boasts shorter model training and updating times, making it more suitable for online BCI control when compared to deep learning methods.

The aforementioned results highlight that the TFSG features, along with the proposed feature selection and classification models in this paper, can effectively enhance the performance of motor imagery EEG decoding. Subsequently, a deeper analysis and discussion of these two aspects of the work are undertaken.

Table 7 Comparison of the classification accuracy with existing deep learning methods (unit: %)

Assessment methods	Datasets	Deep ConvNet Schirrmeister et al. (2017)	EEGNet-8,2 Lawhern et al. (2018)	Spectral-Spatial CNN Kwon et al. (2020)	MIN2NET Autthasan et al. (2021)	MTF-SG(PLV + DE) + LOG	MTF-SG(PLV + DE) + LASSO-F
Subject-dependent	Dataset 1 (L vs R)	63.72 ± 17.18	65.93 ± 18.44	76.91 ± 13.75	65.23 ± 16.14	79.55 ± 15.05	**79.55 ± 14.67**
	Dataset 2	61.40 ± 15.66	67.76 ± 18.09	76.76 ± 16.66	65.90 ± 16.50	**78.45 ± 13.75**	78.33 ± 14.17
Subject-independent	Dataset 1 (L vs R)	56.34 ± 8.86	64.26 ± 11.03	66.05 ± 13.70	60.03 ± 9.24	70.76 ± 13.90	**71.45 ± 13.48**
	Dataset 2	65.26 ± 16.83	58.07 ± 11.45	66.21 ± 15.15	59.79 ± 13.72	**68.57 ± 16.90**	68.33 ± 15.85

Bold text indicates the method with the highest classification accuracy in the target dataset

Impact of feature fusion methods on classification accuracy

In the preceding experimental results, the classification outcomes of TFSG features, comprising PLV and node degree graph features, were exclusively provided. In this section, a detailed discussion on the impact of various brain network construction methods and diverse types of graph features on EEG classification performance is undertaken. Owing to space constraints, the focus will be on the subject-dependent case.

We initially explore the classification results of the MTF-SG method using a single type of graph feature. Figure 4 displays these results using a heatmap, where the vertical coordinates represent eight types of graph features, and the horizontal coordinates represent the three feature selection and classification methods. The values in the figure indicate the average classification accuracy across all data. Key conclusions can be drawn from the classification results in Fig. 4. Firstly, graph features based on the PLV-based brain network demonstrate superior classification results compared to the other two brain network construction methods. Secondly, the classification outcomes of LOG and LASSO-F surpass those of the LASSO method.

The classification results of the MTF-SG method with two types of graph features are depicted in Fig. 5. In this figure, only the combination of global and local features is considered, where the vertical coordinates represent global features, and the horizontal coordinates represent local features. Observing the classification results in Fig. 5, it becomes apparent that the combination of PLV with the proposed LOG and LASSO-F methods yields superior classification outcomes. Despite achieving the highest average classification accuracy of 82.95% in Fig. 5, the integration of two types of graph features did not result in a significant improvement in classification performance.

In conclusion, it suffices to focus on a single type of graph feature during time–frequency-spatial-graph feature extraction. Node degree features demonstrate superior performance for classification, while the type of graph features does not significantly impact the classification outcome. Moreover, graph features extracted by the PLV-based brain network construction may possess greater discriminative capabilities.

Working mechanism of feature selection methods

In this section, a deeper exploration of the operational mechanisms of feature selection and classification methods is undertaken, with a specific emphasis on robustness,

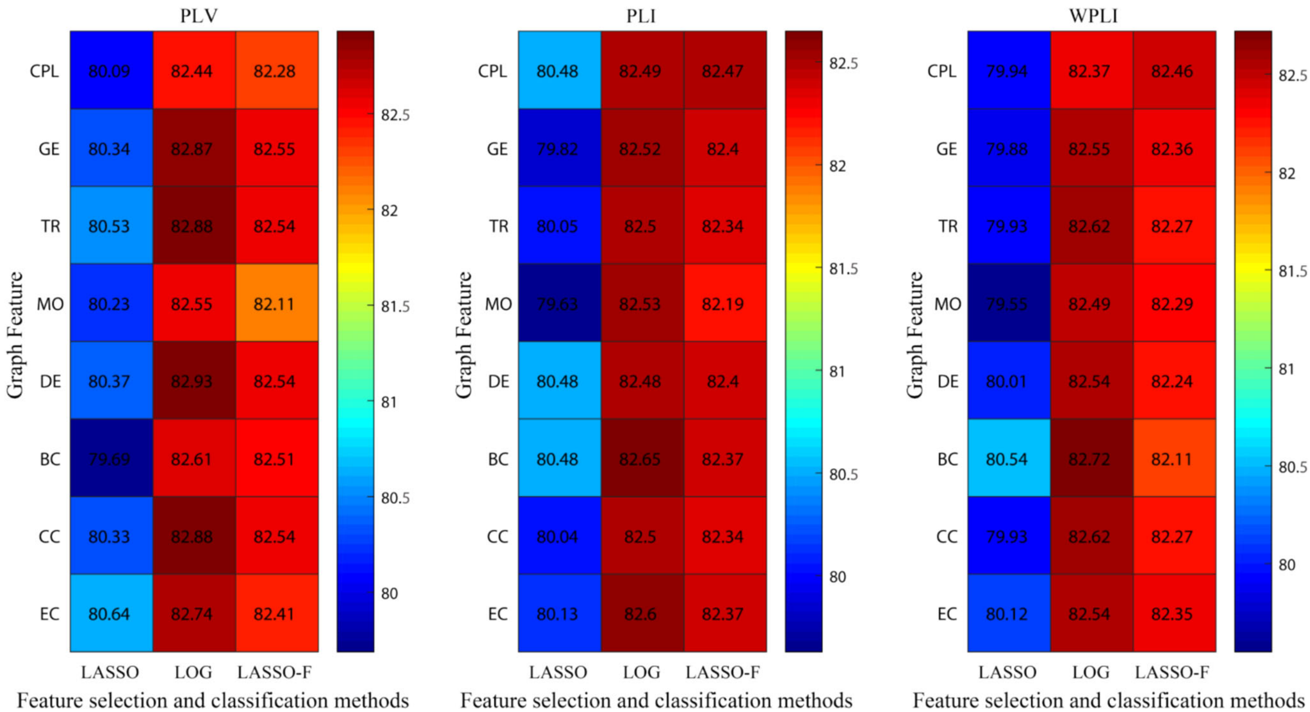

Fig. 4 Average classification accuracy of the MTF-SG method with a single type of graph features (unit: %)

feature separability, and model training time. Similar to the discussion on feature extraction methods, our analysis will be confined to the subject-dependent case.

The robustness of the feature selection and classification models is assessed by examining the weight coefficients **w** of the models. Taking subject A06_FT (FT denotes foot vs tongue binary classification task) as an example, and utilizing the feature extraction method MTF-SG(PLV + DE), the feature weight distribution of the feature selection and classification models is depicted in Fig. 6. From Fig. 6, it becomes evident that the feature weight distribution of the LASSO model is highly scattered, and the variation in the weight values is substantial, making it susceptible to noise interference. In other words, minor data perturbations can lead to significant changes in the model output. In contrast, the LOG and LASSO-F models exhibit a smoother feature weight distribution with less variation, indicating that the models are less sensitive to noise. Consequently, the robustness of LOG and LASSO-F is superior.

To delve deeper into feature separability, the relationship between classification accuracy and Fisher's ratio (Eq. (3–8)) for the feature selection and classification methods is examined. Utilizing subject A06_FT as an example and employing the feature extraction method MTF-SG(PLV + DE), the results are summarized in Table 8. From Table 8, it becomes apparent that there exists a clear correspondence between classification accuracy and Fisher's ratio; namely, a higher classification accuracy corresponds to a larger Fisher's ratio. The two proposed feature selection and classification methods exhibit a significant enhancement in the Fisher's ratio, leading to a notable improvement in classification accuracy.

Finally, the model training time of the three feature selection and classification methods was delved into. The program ran in the following environment: OS: Windows 10, CPU: Intel(R) Core(TM) i5-7300HQ @2.50GHz, RAM: 8GB, MATLAB R2018a. The model training time is measured in seconds, and the results in Fig. 7 are averaged over all subjects in each dataset. Additionally, the model training time for each subject is provided in Tables S7 and S8 of the Supplementary file. From Fig. 7, it is evident that the model training time for LOG and LASSO-F is significantly reduced, especially for the LOG model, irrespective of the feature extraction method.

In summary, it is evident that sparse optimization models based on nonconvex or Fisher's criterion regularization contribute to enhancing EEG decoding performance. The LOG and LASSO-F models, in particular, prove to be user-friendly and require less time for model training.

Future work

In this paper, we observed that graph features extracted by PLV-based brain networks demonstrated greater effectiveness. However, our comparison included only three brain network construction methods, and further

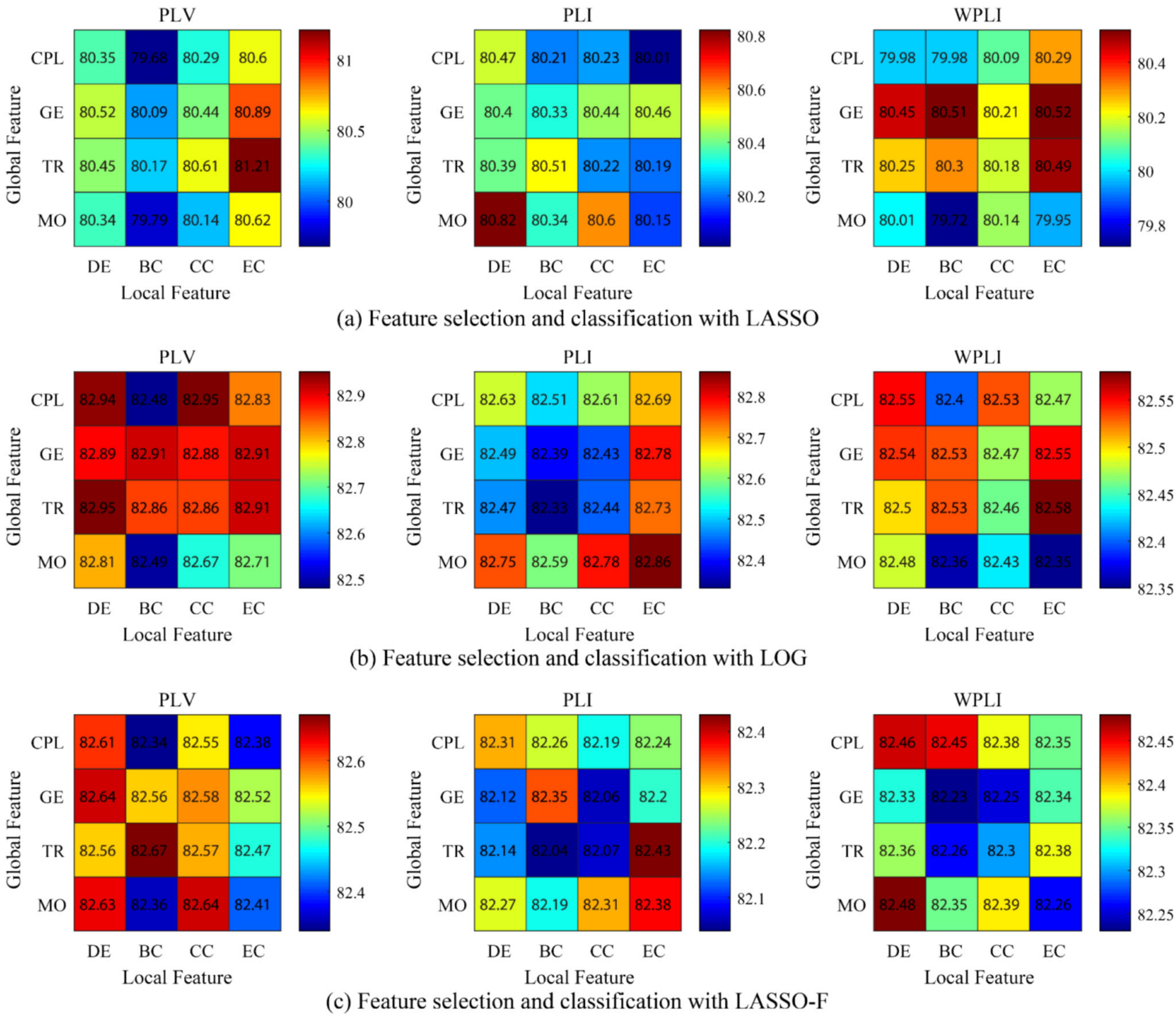

Fig. 5 Average classification accuracy of the MTF-SG method with two types of graph features (unit: %)

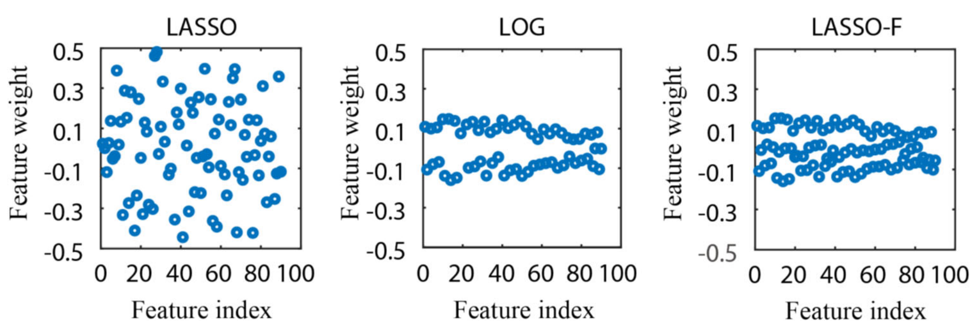

Fig. 6 Feature weight distribution of subject A06_FT. The feature weight distribution is used for evaluating the robustness of the three feature selection and classification models

Table 8 Classification accuracy and Fisher ratio of subject A06_FT

Evaluation index	LASSO	LOG	LASSO-F
Accuracy	70.14	79.86	81.94
Fisher ratio	0.5976	1.3239	1.3805

exploration is warranted, including the investigation of alternative methods such as incorporating Granger causality with directional information. The proposed LOG and LASSO-F models, characterized by their simplicity and

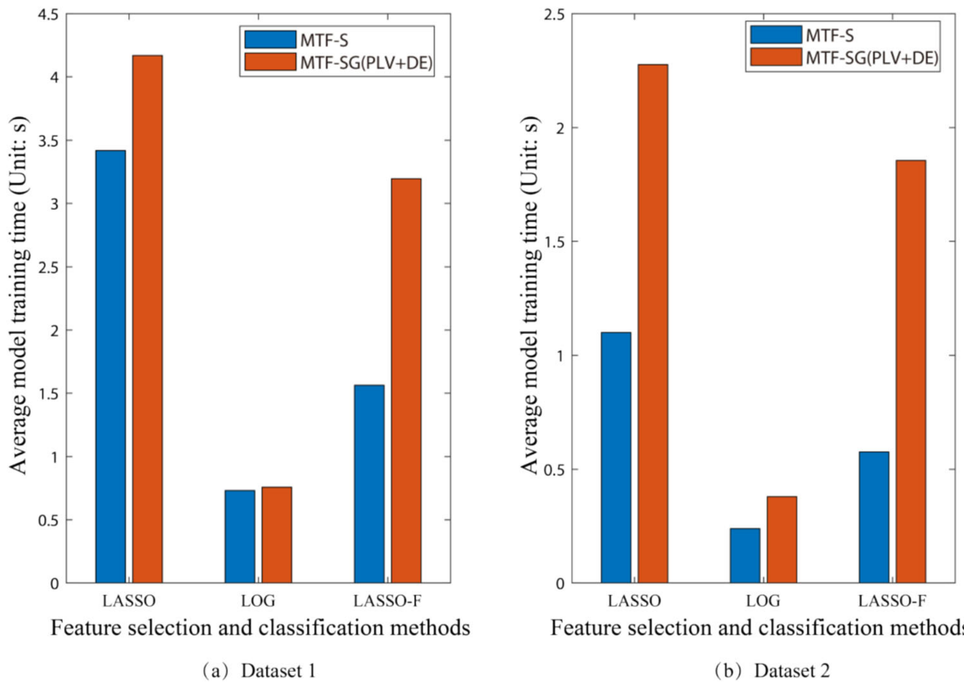

Fig. 7 Average model training time for all subjects in each dataset

ease of use, will be extended for application in other EEG analyses to further validate their effectiveness.

Moreover, the fast-paced evolution of deep learning has prompted us to explore the incorporation of graph information within the existing deep learning framework. One avenue we are exploring involves fusing graph information extracted using graph convolution networks with the time–frequency-spatial information extracted by convolutional neural networks. This approach enables end-to-end EEG decoding.

Conclusion

In this paper, graph-based feature learning methods for subject-dependent and subject-independent motor imagery EEG decoding are presented. Initially, TFSG fused features are extracted to create a comprehensive EEG feature space that is more adept at handling intra-individual variations and inter-individual differences in EEG signals. Subsequently, two feature selection models incorporating non-convex regularization and Fisher's criterion are proposed to facilitate the learning of more precise and separable EEG patterns, resulting in a more discriminative TFSG feature space. Experimental results across subject-dependent, subject-independent, and subject-adaptive scenarios validate the effectiveness of the TFSG feature extraction and discriminative feature selection methods. The proposed feature extraction and selection techniques offer a novel approach for decoding motor imagery EEG. Notably, the LOG and LASSO-F models, though simple, are highly effective and have the potential to be extended to other engineering fields.

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Data availability In this paper, we used two datasets for experiments. Datasets 1 and 2 are public EEG data sets, which have been deposited on the official website.

References

Ai Q, Chen A, Chen K et al (2019) Feature extraction of four-class motor imagery EEG signals based on functional brain network. J Neural Eng 16(2):026032

Altaheri H, Muhammad G, Alsulaiman M et al (2023) Deep learning techniques for classification of electroencephalogram (EEG) motor imagery (MI) signals: a review. Neural Comput Appl 35(20):14681–14722

Ang K K, Chin Z Y, Zhang H, et al. (2008) Filter bank common spatial pattern (FBCSP) in brain-computer interface. In: 2008

IEEE international joint conference on neural networks (IEEE world congress on computational intelligence). IEEE, 2390–2397

Autthasan P, Chaisaen R, Sudhawiyangkul T et al (2021) MIN2net: End-to-end multi-task learning for subject-independent motor imagery EEG classification. IEEE Trans Biomed Eng 69(6):2105–2118

Blankertz B, Tomioka R, Lemm S et al (2008) Optimizing spatial filters for robust EEG single-trial analysis. IEEE Signal Process Mag 25(1):41–56

EEG signal processing and feature extraction (2019) Springer, Singapore.

Fu B, Li F, Ji Y et al (2023) SCDAN: learning common feature representation of brain activation for intersubject motor imagery EEG decoding. IEEE Trans Instrum Meas 72:1–15

Gaur P, Gupta H, Chowdhury A et al (2021) A sliding window common spatial pattern for enhancing motor imagery classification in EEG-BCI. IEEE Trans Instrum Meas 70:1–9

Hoffmann U, Vesin JM, Ebrahimi T et al (2008) An efficient P300-based brain-computer interface for disabled subjects. J Neurosci Methods 167(1):115–125

Jin J, Sun H, Daly I et al (2021a) A novel classification framework using the graph representations of electroencephalogram for motor imagery based brain-computer interface. IEEE Trans Neural Syst Rehabil Eng 30:20–29

Jin J, Xiao R, Daly I et al (2021b) Internal feature selection method of CSP based on L1-norm and dempster-shafer theory. IEEE Trans Neural Netw Learn Syst 32(11):4814–4825

Kwon OY, Lee MH, Guan C et al (2020) Subject-independent brain-computer interfaces based on deep convolutional neural networks. IEEE Trans Neural Netw Learn Syst 31(10):3839–3852

Lawhern VJ, Solon AJ, Waytowich NR et al (2018) EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces. J Neural Eng 15(5):056013

Leng J, Yu X, Wang C, et al. (2024) Functional connectivity of EEG motor rhythms after spinal cord injury. Cognitive Neurodyn, 1–15

Li M, Wang Y, Zhu X et al (2020a) A wrapped time-frequency combined selection in the source domain. Biomed Signal Process Control 57:101748

Li S, Chen H, Wang M et al (2020b) Slime mould algorithm: a new method for stochastic optimization. Futur Gener Comput Syst 111:300–323

Li J, Li Y, Du M (2023) Comparative study of EEG motor imagery classification based on DSCNN and ELM. Biomed Signal Process Control 84:104750

Liu G, Tian L, Zhou W (2022) Multiscale time-frequency method for multiclass motor imagery brain computer interface. Comput Biol Med 143:105299

Lotte F, Guan C (2011) Regularizing common spatial patterns to improve BCI designs: unified theory and new algorithms. IEEE Trans Biomed Eng 58(2):355–362

Lu B, Wang F, Wang S et al (2024) Improvement of motor imagery electroencephalogram decoding by iterative weighted Sparse-Group Lasso. Expert Syst Appl 238:122286

Luo T (2023) Parallel genetic algorithm based common spatial patterns selection on time–frequency decomposed EEG signals for motor imagery brain-computer interface. Biomed Signal Process Control 80:104397

Meng M, Yin X, She Q et al (2021) Sparse representation-based classification with two-dimensional dictionary optimization for motor imagery EEG pattern recognition. J Neurosci Methods 361:109274

Meng M, Xu B, Ma Y, et al. (2024) STGAT-CS: spatio-temporal-graph attention network based channel selection for MI-based BCI. Cognitive Neurodyn, 1–16

Miao M, Zeng H, Wang A et al (2017) Discriminative spatial-frequency-temporal feature extraction and classification of motor imagery EEG: an sparse regression and Weighted Naïve Bayesian Classifier-based approach. J Neurosci Methods 278:13–24

Miao Y, Jin J, Daly I et al (2021) Learning common time-frequency-spatial patterns for motor imagery classification. IEEE Trans Neural Syst Rehabil Eng 29:699–707

Mirjalili S, Lewis A (2016) The whale optimization algorithm. Adv Eng Softw 95:51–67

Phadikar S, Sinha N, Ghosh R (2023) Unsupervised feature extraction with autoencoders for EEG based multiclass motor imagery BCI. Expert Syst Appl 213:118901

Rubinov M, Sporns O (2010) Complex network measures of brain connectivity: uses and interpretations. Neuroimage 52(3):1059–1069

Sadiq MT, Yu X, Yuan Z et al (2021) Toward the development of versatile brain-computer interfaces. IEEE Trans Artif Intell 2(4):314–328

Schirrmeister RT, Springenberg JT, Fiederer LDJ et al (2017) Deep learning with convolutional neural networks for EEG decoding and visualization. Hum Brain Mapp 38(11):5391–5420

Shamsi F, Haddad A, Najafizadeh L (2021) Early classification of motor tasks using dynamic functional connectivity graphs from EEG. J Neural Eng 18(1):016015

Sharma N, Sharma M, Singhal A, et al. (2023) Recent trends in EEG based motor imagery signal analysis and recognition: a comprehensive review. IEEE Access

Steyrl D, Scherer R, Faller J et al (2016) Random forests in non-invasive sensorimotor rhythm brain-computer interfaces: a practical and convenient non-linear classifier. Biomed Eng Biomed Technik 61(1):77–86

Sun J, Li C, Wang Z et al (2023) A memristive fully connect neural network and application of medical image encryption based on central diffusion algorithm. IEEE Trans Industr Inf 20(3):3778–3788

Sun J, Yue Y, Wang Y, et al. (2024) Memristor-based operant conditioning neural network with blocking and competition effects. IEEE Transactions on Industrial Informatics

Tang Y, Zhao Z, Zhang S, et al. (2022) Motor imagery EEG decoding based on new spatial-frequency feature and hybrid feature selection method. Math Problems Eng, 2022

Wang S, Selesnick I, Cai G et al (2018) Nonconvex sparse regularization and convex optimization for bearing fault diagnosis. IEEE Trans Industr Electron 65(9):7332–7342

Wang M, Hu J, Abbass HA (2020) BrainPrint: EEG biometric identification based on analyzing brain connectivity graphs. Pattern Recogn 105:107381

Wen F, Chu L, Liu P et al (2018) A survey on nonconvex regularization-based sparse and low-rank recovery in signal processing, statistics, and machine learning. IEEE Access 6:69883–69906

Wu C, Wang Y, Qiu S, et al. (2024) A bimodal deep learning network based on CNN for fine motor imagery. Cognitive Neurodyn, 1–14

Yang W, Wang K, Zuo W (2012) Neighborhood component feature selection for high-dimensional data. J Comput 7(1):161–168

Zhang R, Li X, Wang Y et al (2019) Using brain network features to increase the classification accuracy of MI-BCI inefficiency subject. IEEE Access 7:74490–74499

Zhang B, Zhu G, Zhu Z (2020a) A TV-log nonconvex approach for image deblurring with impulsive noise. Signal Process 174:107631

Zhang S, Zhu Z, Zhang B et al (2020b) The CSP-based new features plus non-convex log sparse feature selection for motor imagery EEG classification. Sensors 20(17):4749

Zhang S, Zhu Z, Zhang B et al (2021) Fused group lasso: a new EEG classification model with spatial smooth constraint for motor imagery-based brain–computer interface. IEEE Sens J 21(2):1764–1778

Zhang S, Zhu Z, Zhang B et al (2022) Overall optimization of CSP based on ensemble learning for motor imagery EEG decoding. Biomed Signal Process Control 77:103825

Zheng M, Yang B, Gao S et al (2021) Spatio-time-frequency joint sparse optimization with transfer learning in motor imagery-based brain-computer interface system. Biomed Signal Process Control 68:102702